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Adaptive Long-Range LoRaWAN Optimization for Real-Time Agricultural Sensor Telemetry

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ABSTRACT

Large-scale precision agriculture requires multi-kilometer wireless telemetry while respecting duty-cycle regulations and battery limits. We formulate an adaptive spreading factor and transmission power optimization protocol, AgriLoRa-Opt, for heterogeneous farmland topologies. The algorithm leverages environmental humidity and canopy density measurements to predict RF path loss and dynamically assign orthogonal channels. Across a 2,500-hectare agricultural pilot, AgriLoRa-Opt expanded network coverage by 42% while doubling sensor battery lifespan to over seven years. However, simultaneous rain-induced signal attenuation across large regions causes packet collisions due to synchronized spreading-factor escalations.

Keywords— Internet of Things, LoRaWAN, Agricultural Monitoring, Wireless Protocols

INTRODUCTION

Agricultural sensor grids span broad rural acreage where cellular connectivity is nonexistent and standard WiFi is range-limited. LoRaWAN provides sub-GHz long-range links, but uncoordinated spreading factor selection leads to catastrophic packet collisions.

METHODOLOGY

AgriLoRa-Opt introduces weather-aware link budgeting into the LoRa Network Server. Nodes receive predictive downlink configuration packets adjusting transmit power and spreading factors before weather events impact path loss.

RESULTS

Deployed over 2,500 hectares of vineyard and grain fields, the system achieved 98.7% packet reception at distances up to 14 km, while extending mote battery life to 7.2 years on a single AA lithium cell.

LIMITATIONS

During sudden heavy downpours, widespread signal fading caused hundreds of nodes to simultaneously request higher spreading factors, temporarily saturating gateway channels and dropping 18% of non-critical telemetry packets.

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